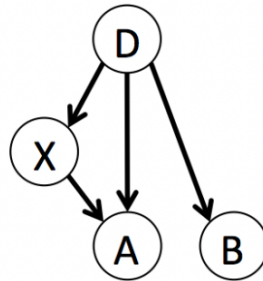


Practice Problem Bayesian Inference

Question 1:

$P(A D, X)$			
+d	+x	+a	0.9
+d	+x	-a	0.1
+d	-x	+a	0.8
+d	-x	-a	0.2
-d	+x	+a	0.6
-d	+x	-a	0.4
-d	-x	+a	0.1
-d	-x	-a	0.9



$P(D)$	
+d	0.1
-d	0.9

$P(X D)$		
+d	+x	0.7
+d	-x	0.3
-d	+x	0.8
-d	-x	0.2

$P(B D)$		
+d	+b	0.7
+d	-b	0.3
-d	+b	0.5
-d	-b	0.5

1: Find the probability of having disease D and getting a positive result on test A?

$$\begin{aligned}
 P(+d, +a) &= p(+d)p(+x|+d)p(+a|+x, +d) + p(+d)p(-x|+d)p(+a|-x, +d) \\
 &= 0.1 * 0.7 * 0.9 + 0.1 * 0.3 * 0.8 = 0.087
 \end{aligned}$$

2: Find the probability of not having disease D and getting a positive result on test A?

$$\begin{aligned}
 P(-d, +a) &= p(-d)p(+x|-d)p(+a|+x, -d) + p(-d)p(-x|-d)p(+a|-x, -d) \\
 &= 0.45
 \end{aligned}$$

3: Find the probability of having disease D given a positive result on test A?

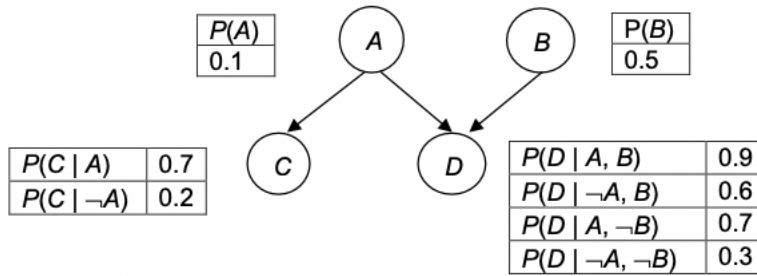
$$P(+d | +a) = 0.087 / (0.087 + .45)$$

4: Find the probability of not having disease D given a positive result on test A?

$$P(-d | +a) = 0.45 / (0.087 + .45)$$

Question 2

Consider the following Bayesian Network containing four Boolean random variables.



1. Compute $P(\neg A, B, \neg C, D)$

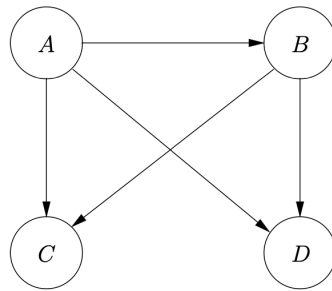
$$P(\neg A, B, \neg C, D) = P(\neg C | \neg A) P(D | \neg A, B) P(\neg A) P(B) \\ = (1 - 0.2) (0.6) (1 - 0.1) (0.5) = 0.216$$

2. Compute $P(A | B, C, D)$

$$P(A | B, C, D) = P(A, B, C, D) / P(B, C, D) \\ P(A, B, C, D) = (0.7) (0.9) (0.1) (0.5) = 0.0315 \\ P(B, C, D) = P(A, B, C, D) + P(\neg A, B, C, D) \\ = 0.0315 + (0.2) (0.6) (0.9) (0.5) = 0.0855 \\ \text{So, } P(A | B, C, D) = 0.0315 / 0.0855 = 0.368$$

Question 3

Consider the following Bayesian network:



The associated probability distributions for the binary random variables A , B , C and D are $\Pr(a) = 0.2$, $\Pr(\neg a) = 0.8$ and:

A	$\Pr(b A)$	A	B	$\Pr(c A, B)$	$\Pr(d A, B)$
$\perp$	0.8	$\perp$	$\perp$	0.4	0.9
$\top$	0.7	$\perp$	$\top$	0.2	0.8
		$\top$	$\perp$	0.3	0.1
		$\top$	$\top$	0.1	0.2

- Write down an expression for the full joint distribution of the random variables A , B , C and D . Compute the probability that A , B and C are $\perp$ while D is $\top$. [Note $\perp$ stands for false, and $\top$ stands for true]

$$P(A,B,C,D) = P(A)P(B|A)P(C|A,B)P(D|A,B)$$

$$P(\perp, \perp, \perp, \top) = P(\sim a)P(\sim b|\sim a)P(\sim c|\sim a, \sim b)P(d|\sim a, \sim b) = 0.8*0.2*0.6*0.9 = 0.0864$$

- Compute the probability distribution of C conditional on the evidence that $D = \perp$.

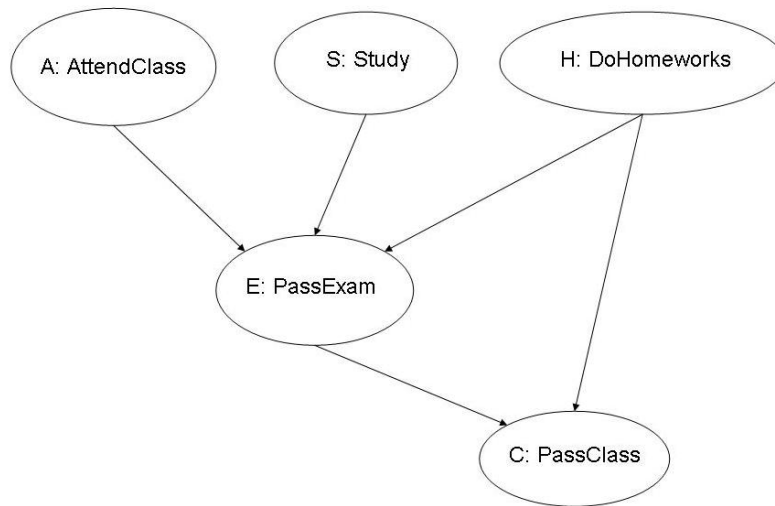
$$\begin{aligned}
 P(c|\sim d) &= \alpha P(a)P(b|a)P(c|a,b)P(\sim d|a,b) + P(a)P(\sim b|a)P(c|a,\sim b)P(\sim d|a,\sim b) + \\
 &\quad P(\sim a)P(b|\sim a)P(c|\sim a,b)P(\sim d|\sim a,b) + P(\sim a)P(\sim b|\sim a)P(c|\sim a,\sim b)P(\sim d|\sim a,\sim b) \\
 &= \alpha 0.2*0.7*0.1*0.8 + 0.2*0.3*0.3*0.9 + 0.8*0.8*0.2*0.9 + 0.8*0.2*0.4*0.1 = \alpha 0.149
 \end{aligned}$$

$$\begin{aligned}
 P(\sim c|\sim d) &= \alpha 0.2*0.7*0.9*0.8 + 0.2*0.3*0.7*0.9 + 0.8*0.8*0.8*0.9 + 0.8*0.2*0.6*0.1 \\
 &= \alpha 0.609
 \end{aligned}$$

$$P(c|\sim d) = 0.149 / (0.149 + 0.609) = .20$$

$$P(\sim c|\sim d) = 0.609 / (0.149 + 0.609) = .80$$

Question 4



1. Write down the expression for the joint distribution as it factorizes according to the Bayes Net below.

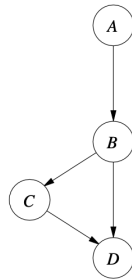
$$P(A,S,H,E,C)=P(A)P(S)P(H)P(E|A,S,H)P(C|E,H)$$

2. Write down the expression for the probability of passing the class, given that you attend class and study but don't do the homeworks.

$$\begin{aligned}
 P(C|s,a,\sim h) &= P(C,s,a,\sim h) / P(s,a,\sim h) \\
 &= \alpha [P(C,s,a,\sim h,e) + P(C,s,a,\sim h,\sim e)] \\
 &= \alpha [P(a)P(s)P(\sim h)P(e|a,s,\sim h)P(C|e,\sim h) + P(a)P(s)P(\sim h)P(\sim e|a,s,\sim h)P(C|\sim e,\sim h)]
 \end{aligned}$$

Question 5

Consider the following Bayesian Network:



The associated probability distributions for the binary random variables A , B , C and D are $\Pr(a) = 0.7$, $\Pr(\neg a) = 0.3$ and:

A	$\Pr(b A)$	B	$\Pr(c B)$	B	C	$\Pr(d B, C)$
$\top$	0.1	$\top$	0.2	$\top$	$\top$	0.6
$\perp$	0.15	$\perp$	0.95	$\top$	$\perp$	0.5
				$\perp$	$\top$	0.4
				$\perp$	$\perp$	0.3

1. Write down an expression for the full joint distribution of the random variables A , B , C and D . Compute the probability that A and B are $\top$ while C and D are $\perp$.
2. Compute the probability distribution of B conditional on the evidence that $D = \perp$.